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No additional materials are needed.



1 Fig. 1.1 shows a coil made from resistance wire.

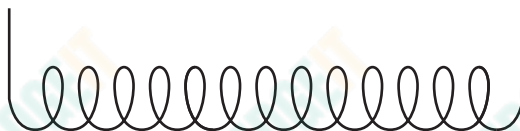


Fig. 1.1

The coil is placed in cooking oil of mass m . The total length of the resistance wire in the oil is L .

A potential difference V is applied to the coil. The temperature of the oil increases by $\Delta\theta$ in time t .

It is suggested that $\Delta\theta$ is related to L by the relationship

$$\frac{AtV^2}{L} = mK\Delta\theta + Z$$

where A is the cross-sectional area of the wire, and K and Z are constants.

Plan a laboratory experiment to test the relationship between $\Delta\theta$ and L .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for K and Z .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.





Diagram

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DO NOT WRITE IN THIS MARGIN

- 2 A student investigates the refraction of white light entering a transparent rectangular block. A narrow beam of light enters the block at the midpoint of one of the shorter sides. The angle of incidence θ is measured, as shown in Fig. 2.1.

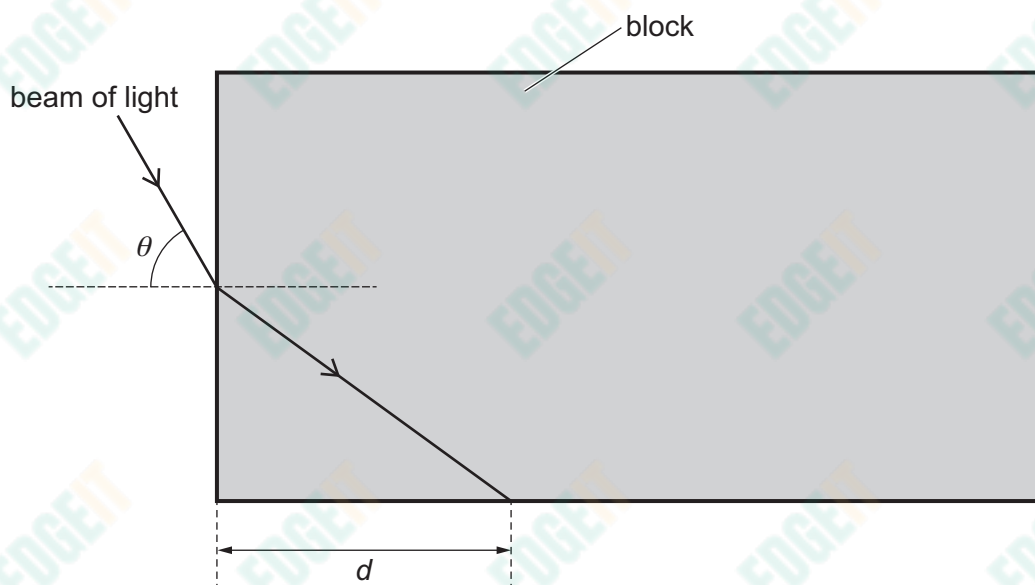


Fig. 2.1 (not to scale)

The distance d between the corner of the block and the point where the beam of light touches the boundary of the block is measured.

The experiment is repeated for different values of θ .

It is suggested that d and θ are related by the equation

$$\frac{B}{B + d^2} = \frac{\sin^2 \theta}{n^2}$$

where B and n are constants.

- (a) A graph is plotted of d^2 on the y -axis against $\frac{1}{\sin^2 \theta}$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]





- (b) Values of θ , $\frac{1}{\sin^2 \theta}$ and d are given in Table 2.1.

Table 2.1

$\theta / ^\circ$	$\frac{1}{\sin^2 \theta}$	d / cm	d^2 / cm^2
28.5	4.39	24.8 ± 0.2	
33.5	3.28	21.4 ± 0.2	
42.5	2.19	17.1 ± 0.2	
50.0	1.70	14.7 ± 0.2	
57.5	1.41	12.9 ± 0.2	
63.5	1.25	11.8 ± 0.2	

Calculate and record values of d^2 / cm^2 in Table 2.1.
Include the absolute uncertainties in d^2 .

[2]

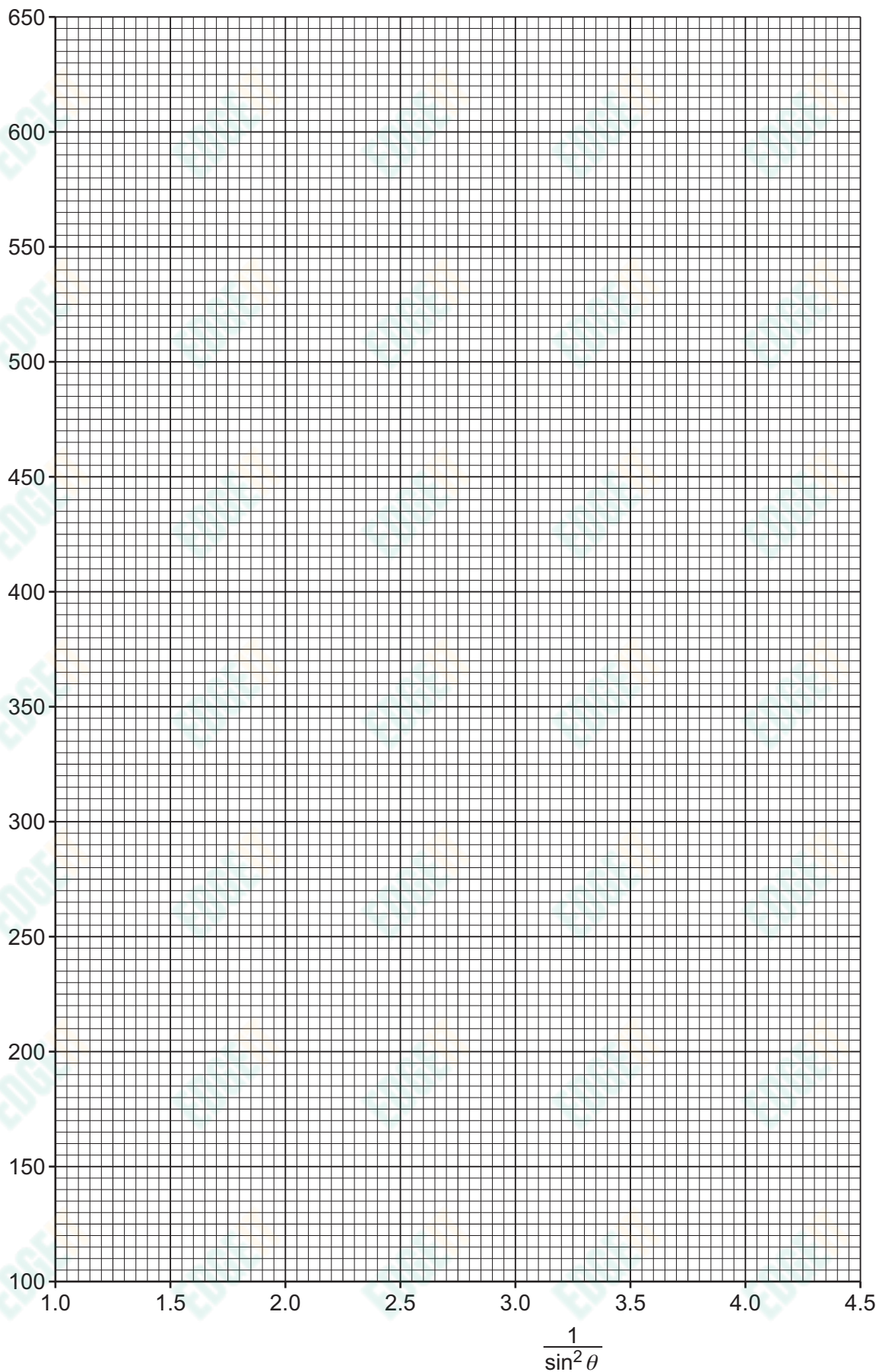
- (c) (i) Plot a graph of d^2 / cm^2 against $\frac{1}{\sin^2 \theta}$. Include error bars for d^2 . [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]





d^2/cm^2





- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of B and n . Include appropriate units.

B =

n = [2]

- (ii) Determine the percentage uncertainty in n .

percentage uncertainty in n = % [1]

- (e) The experiment is repeated. Determine the angle θ that gives a value of d of 30.0 cm.

θ = ° [1]

[Total: 15]

